

Yu Liu

Postdoctoral Fellow
Institute of Advanced Computing and Digital Engineering
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Chinese Academy of Sciences

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Summary

My research interests broadly lie in **Computational Medicine**, focus on imaging computing, signal processing, computational fluid dynamics (CFD) and their application in cardiology, neurology and neuroscience.

Education

The Chinese University of Hong Kong

Ph.D. in Medical Sciences

Hong Kong, China

Aug 2022 - Oct 2025

- GPA: 3.57/4

- Advisor: Prof. Xinyi Leng & Prof. Thomas Leung

- Thesis: Geometric and Hemodynamic Characteristics of Intracranial Atherosclerotic Stenosis: Clinical, Imaging, and Computational Investigations

Beihang University

M.Phil. in Biomedical Engineering

Beijing, China

Sep 2018 - Jun 2021

- GPA: 3.8/4

- Postgraduate recommendation

- Advisor: Prof. Yawei Wang & Prof. Yubo Fan

- Thesis: Hemodynamic Study of Vertebrobasilar Dolichectasia

Hebei University of Technology

B.Eng. in Biomedical Engineering

Tianjin, China

Sep 2014 - Jun 2018

- GPA: 3.71/4

- Honor Graduate with Distinction (Department Rank: 1/41)

- Thesis: Non-contact Heart Rate Detection Based on Photoplethysmographic Imaging (**Best thesis award, 5%**)

Experience

Shenzhen Institutes of Advanced Technology,

Chinese Academy of Sciences

Postdoctoral Fellow

Shenzhen, China

Dec 2025 - Present

- Laboratory for Engineering and Scientific Computing

- Institute of Advanced Computing and Digital Engineering

- Advisor: Prof. Jia Liu

Fosun Aitrox

Research and Development Engineer

Shanghai, China

Jun 2021 - May 2022

- Project: Hemodynamic algorithm for fraction flow reserve computed tomography (FFRct)

China National Institute of Standardization

Research Intern

Beijing, China

Aug 2019 - Oct 2019

- Project: Standardized and digital management of medical insurance policy

Funding

National Natural Science Foundation of China (NSFC), 2021-2024 (Co-Investigator)

- Affiliated institution: Beihang University

Health and Medical Research Fund (HMRF), 2020-2021 (Co-Investigator)

- Affiliated institution: The Chinese University of Hong Kong/Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

National Key Research and Development Plan, Ministry of Science and Technology of the People's Republic of China, 2020-2023 (Co-Investigator)

- Affiliated institution: Sun Yat-sen University/Beihang University

National Natural Science Foundation of China (NSFC)/ Research Grants Council (RGC)

of Hong Kong Joint Research Scheme, 2017-2020 (Co-Investigator)

• Affiliated institution: Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences/The Chinese University of Hong Kong

Publications

Journal

2026

• Wall Shear Stress Distribution in Intracranial Atherosclerotic Disease and Associations With Vessel and Plaque Morphology

Linfang Lan*, Shuang Li*, Haipeng Liu, Vincent H.L. Ip, Jill Abrigo, Xuan Tian, Yuying Liu, **Yu Liu**, Ziqi Li, Tingjun Liang, Florence S. Y. Fan, Sze Ho Ma, Karen Ma, Bonaventure Y.M. Ip, Yannie O. Y. Soo, Howan Leung, Vincent C. T. Mok, Hui Fang, Bo Song, Yuming Xu, Yuhua Fan, Thomas W. Leung, Xinyi Leng[†]

CNS Neuroscience & Therapeutics (CNSNT), 2026

2025

• Enlarged translesional pressure gradient drives recruitment of leptomeningeal collaterals in medically treated patients with symptomatic middle cerebral artery stenosis

Yuying Liu, Xuan Tian, Jill Abrigo, Shuang Li, **Yu Liu**, Linfang Lan, Haipeng Liu, Conaventure YM Ip, Sze Ho Ma, Karen Ma, Florence SY Fan, Hing Lung Ip, Yannie OY Soo, Howan Leung, Vincent CT Mok, Thomas W Leung, Xinyi Leng[†]

Cerebrovascular Diseases (Cerebrovasc Dis), 2025

• Is Invasive Fractional Flow Measurement Accurate in Intracranial Stenosis? A Computational Simulation Study

Yu Liu*, Zhengzheng Yan*, Ziqi Li, Yuying Liu, Sze Ho Ma, Bonaventure YM Ip, Thomas W Leung, Jia Liu[†], Xinyi Leng[†]

Journal of NeuroInterventional Surgery (JNIS), 2025

• More severe cerebral small vessel disease associated with poor leptomeningeal collaterals in symptomatic intracranial atherosclerotic stenosis

Yuying Liu, Shuang Li, Xuan Tian, Jill Abrigo, Bonnie YK Lam, Jize Wei, Lina Zheng, **Yu Liu**, Ziqi Li, Tingjun Liang, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng[†]

Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2025

• Stroke Mechanisms in Intracranial Atherosclerotic Disease: A Modified Classification System and Clinical Implications

Shuang Li, Xuan Tian, Xueyan Feng, Bonaventure Ip, Hing Lung Ip, Jill Abrigo, Lina Zheng, Yuying Liu, **Yu Liu**, Ziqi Li, Tingjun Liang, Karen KY Ma, Florence SY Fan, Sze Ho Ma, Hui Fang, Bo Song, Yuming Xu, Howan Leung, Yannie OY Soo, Vincent CT Mok, Ka Sing Wong, Xinyi Leng[†], Thomas WH Leung[†]

Translational Stroke Research (Transl Stroke Res), 2025

• Clinical implications of haemodynamics in symptomatic intracranial atherosclerotic stenosis by computational fluid dynamics modelling: a systematic review

Yu Liu, Shuang Li, Haipeng Liu, Xuan Tian, Yuying Liu, Ziqi Li, Thomas W Leung, Xinyi Leng[†]

Stroke and Vascular Neurology (SVN), 2025

2024

• Cerebral hemodynamics and stroke risks in symptomatic intracranial atherosclerotic stenosis with internal versus cortical borderzone infarcts: a computational fluid dynamics study

Shuang Li, Xuan Tian, Bonaventure Ip, Xueyan Feng, Hing Lung Ip, Jill Abrigo, Linfang Lan, Haipeng Liu, Lina Zheng, Yuying Liu, **Yu Liu**, Karen KY Ma, Florence SY Fan, Sze Ho Ma, Hui Fang, Yuming Xu, Alexander Y Lau, Howan Leung, Yannie OY Soo, Vincent CT Mok, Ka Sing Wong, Xinyi Leng[†], Thomas W Leung[†]

Journal of Cerebral Blood Flow & Metabolism (JCBFM), 2024

• Hemodynamic significance of intracranial atherosclerotic disease and ipsilateral imaging

markers of cerebral small vessel disease

Lina Zheng, Xuan Tian, Jill Abrigo, Hui Fang, Bonaventure YM Ip, Yuying Liu, Shuang Li, **Yu Liu**, Linfang Lan, Haipeng Liu, Hing Lung Ip, Florence SY Fan, Sze Ho Ma, Karen Ma, Alexander Y Lau, Yannie OY Soo, Howan Leung, Vincent CT Mok, Lawrence KS Wong, Yuming Xu, Liping Liu, Xinyi Leng[†], Thomas W Leung[†]
European Stroke Journal (ESJ), 2024

2023

• Effects of abnormal vertebral arteries and the circle of Willis on vertebrobasilar dolichoectasia: a multi-scale simulation study

Yu Liu^{*}, Xinmiao Zhang^{*}, Yawei Wang[†], Wentao Feng, Jing Jing, Zhujun Sun, Bitian Wang, Yongjun Wang, Yubo Fan[†]
Clinical Biomechanics (Clin Biomech), 2023

2022

• Effects of stent shape on focal hemodynamics in intracranial atherosclerotic stenosis: A simulation study with computational fluid dynamics modeling

Haipeng Liu, **Yu Liu**, Bonaventure Y M Ip, Sze Ho Ma, Jill Abrigo, Yannie O Y Soo, Thomas W Leung, Xinyi Leng[†]
Frontiers in Neurology (Front Neurol), 2022

• Variations of human cerebral and ocular blood flow during exposure to multi-axial accelerations: A mathematical modeling study

Weipeng Li, Bitian Wang, Yawei Wang[†], Xiaoyu Liu, Wentao Feng, Tianya Liu, Zhujun Sun, **Yu Liu**, Songyang Liu, Yubo Fan[†]
Medical & Biological Engineering & Computing (MBEC), 2022

Conference Abstract

• Impact of catheter and pressure wire on the measurement of fractional flow in intracranial atherosclerotic stenosis: a computational simulation study

Yu Liu, Zhengzheng Yan, Ziqi Li, Yuying Liu, Sze Ho MA, Bonaventure Ym Ip, Thomas Leung, Jia Liu, Xinyi Leng

Poster presentation, *European Stroke Organisation Conference (ESOC) 2025*, Helsinki, Finland

• Symptomatic atherosclerotic middle cerebral artery stenosis: associations of proximal arterial geometry with plaque location and focal hemodynamics

Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Ziqi Li, Tingjun Liang, Haipeng Liu, Bonaventure YM Ip, Hui Fang, Thomas W Leung, Xinyi Leng

Poster presentation, *European Stroke Organisation Conference (ESOC) 2025*, Helsinki, Finland

• Associations of ICA Bifurcation Geometry, MCA Geometry and Plaque Location in Symptomatic Atherosclerotic MCA Stenosis

Yu Liu, Shuang Li, Linfang Lan, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure Ym Ip, Thomas Leung, Xinyi Leng

Poster presentation, *European Stroke Organisation Conference (ESOC) 2024*, Basel, Switzerland

• Association between Adjacent Arterial Geometry and Plaque Features in Symptomatic Atherosclerotic Middle Cerebral Artery Stenosis

Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Ziqi Li, Haipeng Liu, Bonaventure YM Ip, Thomas W Leung, Xinyi Leng

Poster presentation, *Asia Pacific Stroke Conference (APSC) 2023*, Hong Kong, China

• Cerebral hemodynamics by computational fluid dynamics modeling in symptomatic intracranial atherosclerotic disease and the clinical relevance: a systematic review

Yu Liu, Shuang Li, Xuan Tian, Yuying Liu, Haipeng Liu, Thomas Leung, Xinyi Leng

Poster presentation, *European Stroke Organisation Conference (ESOC) 2023*, Munich, Germany

* indicates equal contribution.
† indicates corresponding authorship.

Challenge First prize, National Biomedical Engineering Innovation Design Competition for College Students (**top 10%**)
Project: Low-powered and remoted electrocardiogram (ECG) monitoring system
Third prize, iCAN Innovation Contest
Project: Intelligent wheelchair based on real-time pose transformation

Honors & Awards	Research postgraduate scholarship, <i>The Chinese University of Hong Kong</i>	2022-2025
	First-class scholarship, <i>Beihang University</i>	2018-2021
	Outstanding graduate, <i>Hebei Province (top 5%)</i>	2018
	Outstanding graduate, <i>Hebei University of Technology</i>	2018
	First-class scholarship, <i>Hebei University of Technology</i>	2017-2018
	Outstanding student leader, <i>Hebei University of Technology</i>	2017
	Outstanding student, <i>Hebei University of Technology</i>	2016

Skills **Programming:** Python, MATLAB, C/C++, LATEX, HTML, R
Libraries & OS: OpenCV, CGAL; Linux (Ubuntu), Mac OSX
Tool: Git, Docker
Languages: Mandarin, English